
Harmonizing Historical Trade Using Geopolitical Data: A Multilayer Network Approach to Bilateral Flows, 1830–1938

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Abstract

Our research project reconstructs historical networks of world bilateral trade from the early stages of the nineteenth-century globalization to the eve of WWII (c. 1830–1938). Its sources are trade tables produced by national customs administrations gathered in the RICardo database (<https://ricardo.medialab.sciences-po.fr/>) (Dedinger and Girard 2017). The collected trade flows (570,000 over 1830-1938) are extremely difficult to work with as they include a total of 2,350 trading entities encompassing not only countries or colonies, but also miscellaneous entities like groups of countries, cities or colonial and geographical areas. 22% of the total bilateral flows (average value share over the entire period) involve at least one such entity. Drawing on the transdisciplinary skills of our team, we have developed a method to reduce this diversity and build a trade network fit for quantitative analysis.

We build yearly multilayer networks by combining trade flows from the RICardo dataset with geopolitical links between trading entities from the GeoPolHist dataset (<http://medialab.github.io/GeoPolHist>) (Dedinger and Girard 2021) (figure 1). Using the geopolitical layer, we first impute transformations converting trading entities into autonomous political entities. In a second step, we apply those transformations by generating new flows from the aggregations or splits of the original trade flows. Splitting trade values into smaller parts is achieved by looking for ratios in time-adjacent trade networks or by imputing ratios with a gravity model (figure 2).

We convert into bilateral flows between countries or colonies 75% of the bilateral flows corresponding to exchanges involving at least one miscellaneous entity (cities, groups of countries, areas), which accounts for 19.4% on average of the total value of bilateral flows over the period. The "aggregation", "split by ratio in adjacent years" and "gravity" methods solved 13%, 2.7% and 3.6% of this amount respectively.

By transforming the very heterogenous trade partner nomenclatures from the sources into one unified system we build a more robust trade network for quantitative analysis. Our

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method yields an average 156% increase in trade network density across the period. The average density grows from 1.6% to 4.3%, including trade flows and partners we could not solve (figure 3).

Our method greatly improves the calculability of the global bilateral trade network, yet it does not eradicate heterogeneity. First, we are missing many potential data sources that could have been produced by trading entities. In many cases, their trade can only be reconstructed from their partners' statistics. Second, we cannot yet break down some large areas (geographical or colonial) or groups with our intentionally limited set of assumptions.

Nevertheless, our reconstructed yearly trade networks provide a new research tool for exploring the causes and effects of trade globalization in the 19th and 20th century. They can be used to track the effects of imperialist expansion that occurred in the long 19th century and draw important conclusions on the impact of global geopolitical shifts on trade structure - a theme notably relevant to the debate surrounding the motivation behind the Scramble for Africa (Frankema, Williamson, and Woltjer 2018).

References

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Keywords: Trade history, Multilayer network, Trade network, Geopolitical history

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Figure 1: trade resolution network for 1850

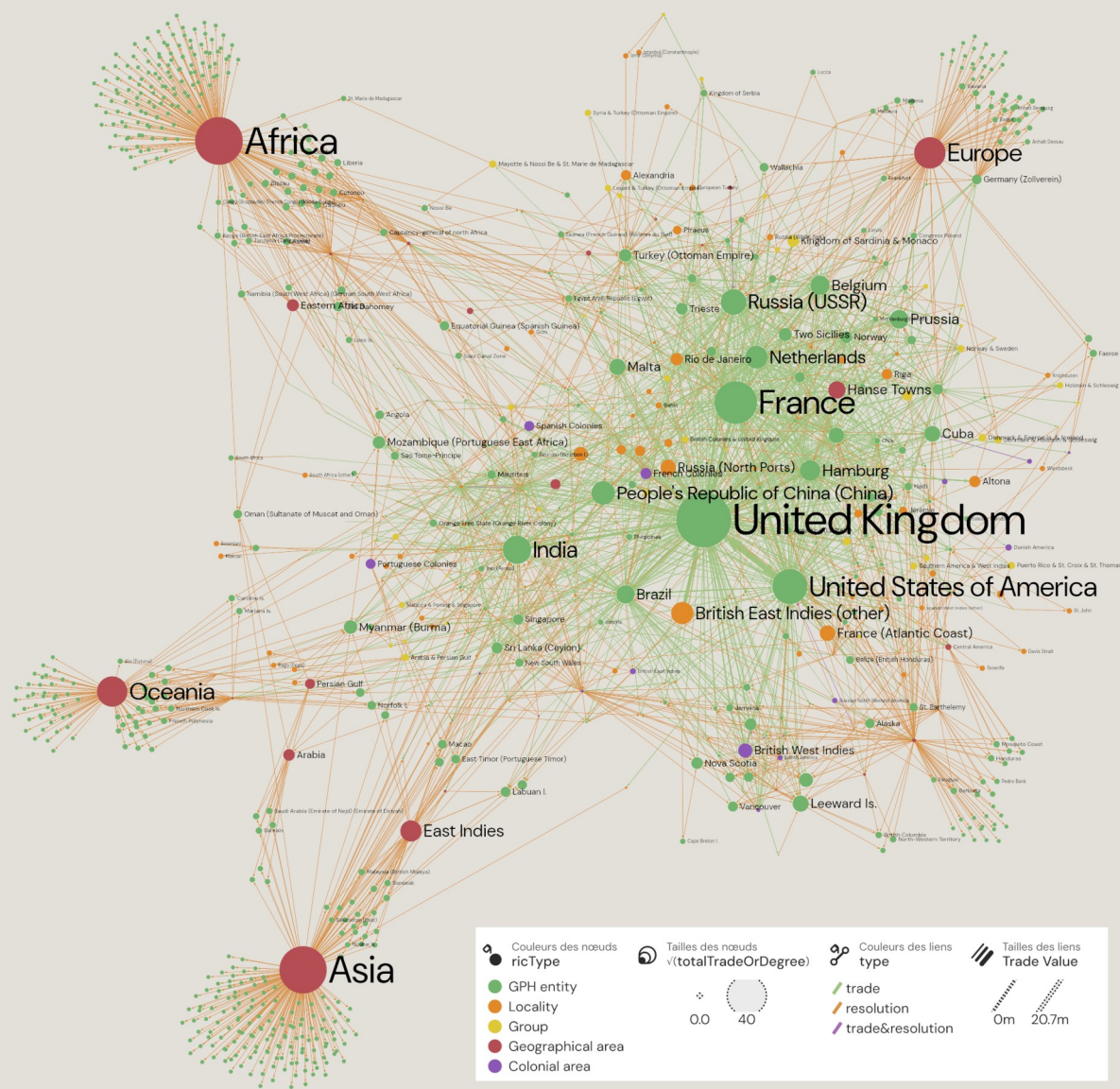


Figure 1.

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Figure 2: trade resolution methods examples extracted from 1850 network

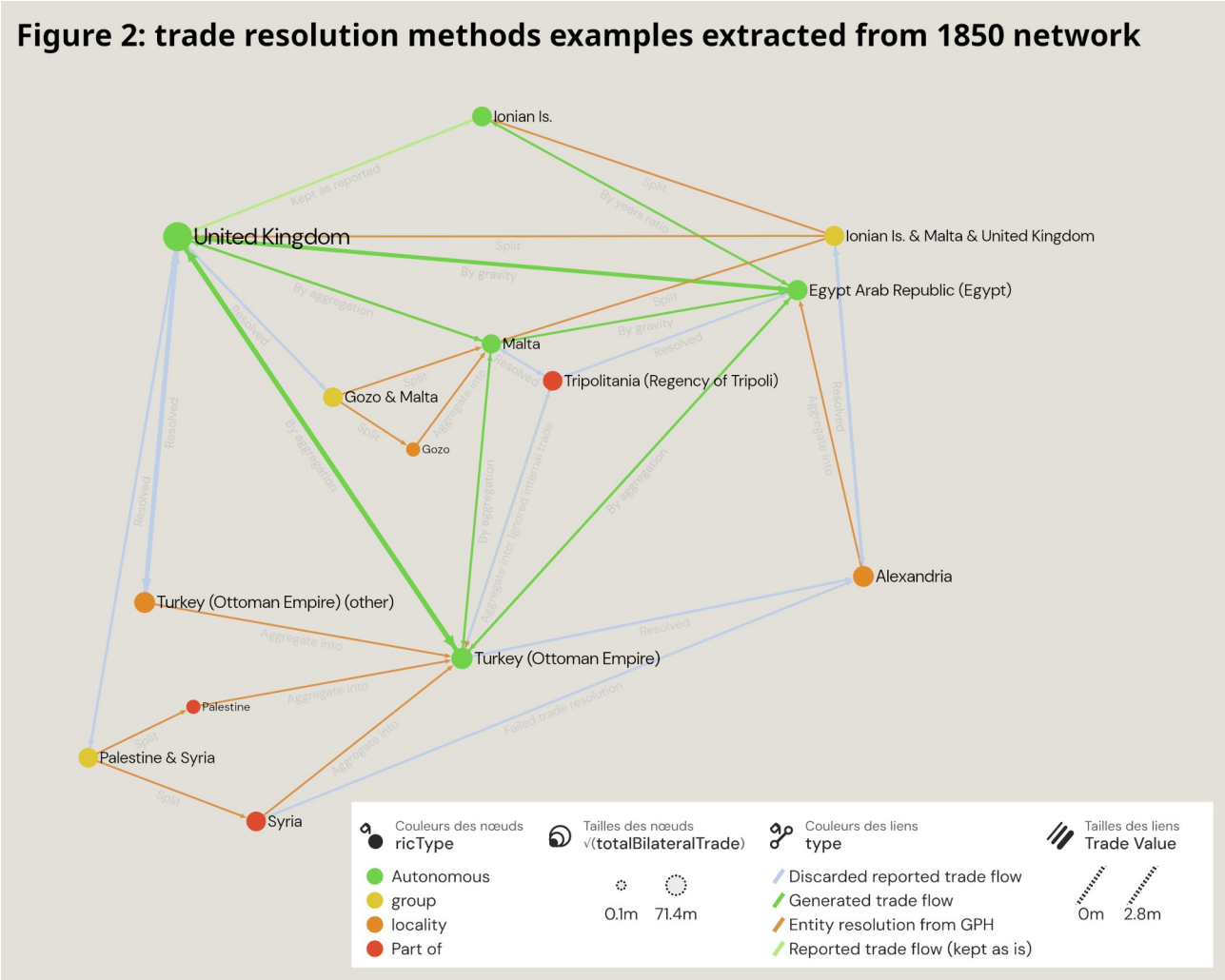


Figure 2.

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Figure 3: Statistical overview of trade resolution applications on RICardo dataset 1833-1938

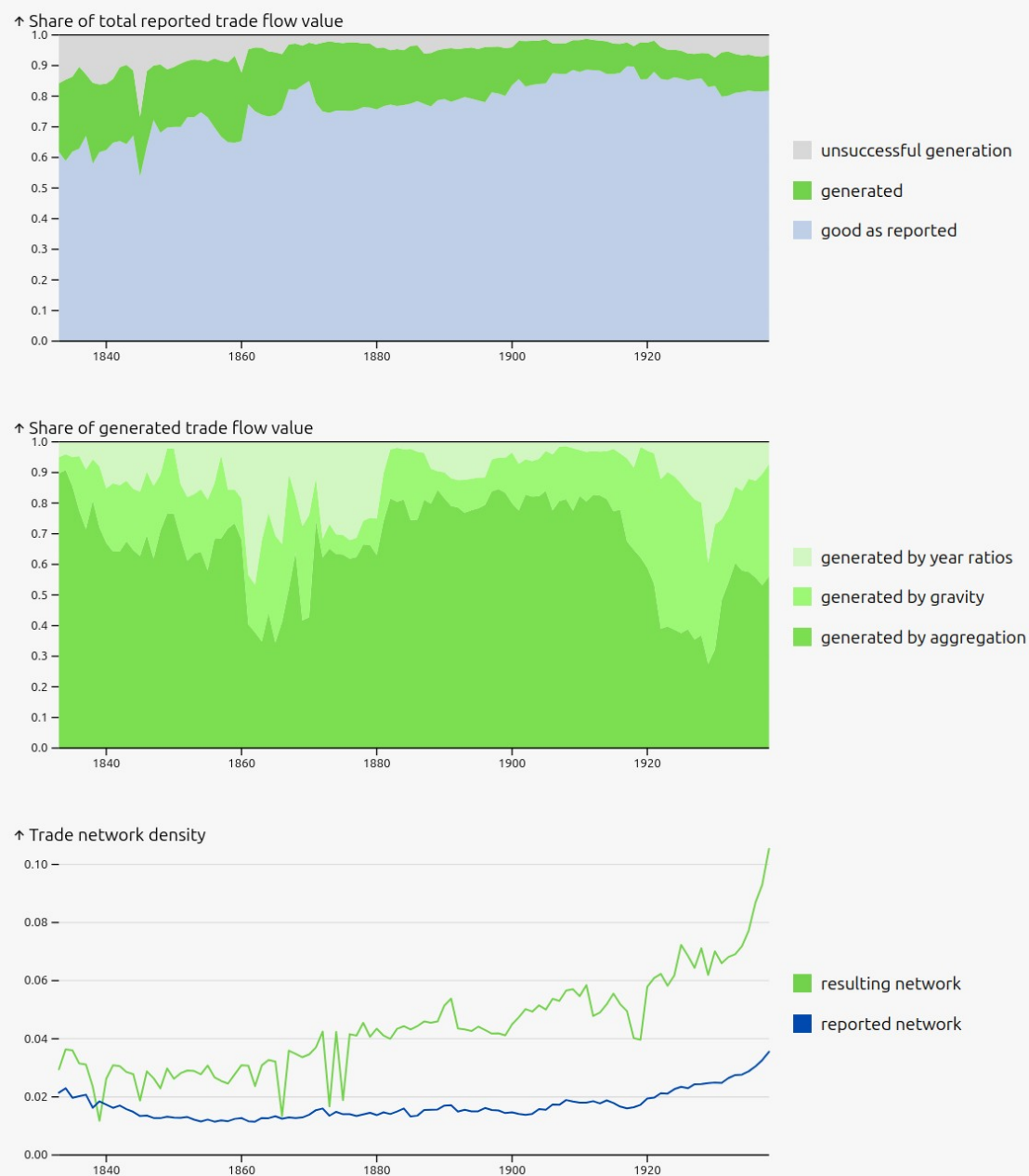


Figure 3.